



PIYUSH GIRISH KULKARNI

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PROFILE SUMMARY

AI & Data enthusiast with hands-on experience engineering both rapid prototypes and production-grade GenAI solutions using state-of-the-art LLMs. Proven track record of bridging technology and business, translating complex AI capabilities into accessible tools, reliable workflows, and actionable client enablement.

SKILLS

- LLM & GenAI Ecosystem: OpenAI (GPT-4 / ChatGPT), Google Gemini, Prompt Engineering, RAG Architectures, LangChain, Agentic Systems, MCP Servers, REST APIs.
- AI Prototyping & Tools: Streamlit, Python (Pandas, NumPy), Power BI, Automation Workflows, Data Visualization.
- Databases & Infrastructure: SQL, PostgreSQL, ChromaDB (Vector DB), Git, AWS, Docker, Airflow, Dagster.

WORK EXPERIENCE

GenAI & Agentic AI Engineer (Company Project) | Galagos AI – Hamburg, Germany | *March 2025 – August 2025*

- Prototyping & Technical Leadership: Led a 5-member team to architect prototype-ready search engines using Google Gemini, LangChain, and ChromaDB, translating complex domain queries into automated workflow and tool recommendations.
- Prompt Engineering & Agent Optimization: Leveraged Gemini APIs within autonomous Self-RAG agent frameworks, designing targeted prompts and evaluation layers to iteratively improve context retrieval accuracy for specialized terminology.
- Technical Enablement: Directed architectural choices for automated data ingestion and platform integrations (via MCP servers and EXA Search), bridging technical implementation with user-centric functionality.

Data Scientist | Quantum Innotech Solutions – India | *Dec 2023 – Aug 2024*

- Algorithm Development: Engineered a targeted search and matching algorithm utilizing KNN and FLAN to process complex input data structures and deliver highly relevant results.
- Backend Intelligence: Built custom relational dictionary databases and data access layers via SQLAlchemy and MySQL, developing Flask backends to manage model operations.
- CI/CD Deployment: Enabled seamless continuous deployment of algorithmic models to Linode production servers by building and maintaining a customized Git-based CI/CD pipeline.

Software Engineer & Data Analyst | CIS IT Solutions Pvt Ltd – India | *Feb 2023 – Dec 2023*

- Predictive Modeling: Developed custom machine learning models using Python and NumPy to significantly enhance client predictive analytics and model forecasting accuracy.
- Data Exploration: Supported epidemiological research by conducting deep exploratory data analysis (EDA) and engineering interactive data visualizations using Power BI and MS Excel.
- Workflow Automation: Streamlined client operations and data collection capabilities by developing automated cloud workflow applications using Microsoft Power Apps and AWS.

Data Engineer (Working Student) | STATISTA GmbH – Hamburg, Germany | *July 2025 – Current*

- AI-Driven Pipeline Automation: Engineered automated ETL/ELT pipelines using Python and REST APIs, integrating OpenAI GPT models for automated metadata extraction, text classification, and dataset structuring across 20+ international economic sources.
- Performance & Quality Optimization: Implemented prompt templates and chunked processing in Pandas to efficiently process millions of raw records while maintaining high data quality standards.

EDUCATION

SRH University of Heidelberg Campus Hamburg, Hamburg, Germany

Master of Science – Applied Data Science & Analytics (Oct 2024 – Present)

- Relevant Coursework: Machine Learning, Deep Learning, Big Data with Python, Statistics, Data Analysis, Data Management.

MIT ADT University, Pune, India

Bachelor of Technology – Computer Science & Engineering (Class of 2022)

- Relevant Coursework: Core AI Technologies, Deep Learning, Machine Learning.

ACADEMIC PROJECTS

FoodVisionAI-Food Recognition & Nutrition App - [Link](#)

- Vision Architecture: Fine-tuned PyTorch CNNs (EfficientNet) on a large-scale dataset of 131K+ images to achieve a 95% real-time dish classification accuracy.
- Semantic Matching Engine: Utilized Sentence Transformers to accurately map visual predictions to a 1,000+ item nutrition database with an ultra-low query latency of ~14ms.
- GenAI Deployment: Built a responsive Streamlit web application integrating local LLMs (Ollama) to deliver interactive AI-driven Q&A and comprehensive food descriptions.

Topic Modelling Using LDA (July 2021 to May 2022) - [Link](#)

- NLP Framework: Utilized Latent Dirichlet Allocation (LDA) frameworks to rapidly process, cluster, and analyze extensive corpora of unstructured biomedical text data.
- Summarization Architecture: Built a specialized question-answering model leveraging the PICO framework for precise context extraction from dense COVID-19 datasets.
- Text Processing Pipelines: Engineered automated regex, text tokenization, and vectorization pipelines to transform raw literature into structured machine-readable formats.

IoT Smoke Detection Data Pipeline- [LINK](#)

- ML Integration: Developed and deployed automated Scikit-learn predictive models via REST APIs to handle real-time classification inference.
- Streaming Infrastructure: Built a live event-streaming and batch pipeline using Apache Kafka, Docker, and Airflow to stream high-frequency IoT sensor metrics.
- Model Observability: Integrated Prometheus and Grafana for live data drift monitoring and achieved 95% automated test coverage across the model environment.

ACHIEVEMENTS

- AWS Certified Cloud Practitioner
- Copyright: [Perspective Approach for Mango Ripening Classifiers](#)

LANGUAGES

- English – Native
- German – Conversational (A2 – B1)



Place: Hamburg, Germany

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